
Green Development

Can countries implement strict environmental policies without harming economic growth?



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1. Introduction

International trade and globalization have been catalysts for economic growth. Gross domestic product is the default measure for the size, health, and growth of an economy. However, when diving into the definition of GDP, one can realize it is far from a perfect measure of a country's economic well-being, and it neglects important features of a country's economy. For instance, we may want to consider poverty, wealth inequality, unemployment, and the environment's health. The prioritization of national economic growth has led to sacrifices in many other areas, including sustainability.

Climate change has become a salient concern in the 21st century, provoked by anthropogenic activity and the burning of fossil fuels. The global temperature is now rising at a rapid pace ($\sim 1.1^{\circ}\text{C}$ since 1850), leading to an increased magnitude of droughts, floods, natural disasters, ecosystem damage, sea level rise, and more adverse effects¹. Consequently, the United Nations introduced an international treaty to address climate change at the annual Climate Change Conference in 2015, held in Paris (COP 21). 195 parties agreed to the following objectives: (1) limiting the world's average temperature rise below 2°C above pre-industrial levels, (2) reducing global greenhouse gas emissions significantly (mainly carbon dioxide), (3) reaching net zero emissions in the second half of the 21st century, and 4) providing financial resources to help developing countries implement climate change mitigation plans².

While most countries recognize the agreement, the United States government has shown inconsistent adherence to its principles. At his 2025 inauguration, President Trump stated his rationale for withdrawing from the agreement a second time, stating such

¹ National Oceanic and Atmospheric Administration. "Climate Change Impacts." <https://www.noaa.gov/education/resource-collections/climate/climate-change-impacts>.

² "The Paris Agreement," Unfccc.int, <https://unfccc.int/process-and-meetings/the-paris-agreement>.

agreements “do not reflect our country's values or our contributions to the pursuit of economic and environmental objectives.”³ Some critics of the agreement, like Trump, express concerns over its hindrance of economic growth.

In our research, we explore whether the relationship between environmental policy enforcement and a reduction in economic growth can be empirically proven. According to Mulder et al. (2021), GDP and CO₂ emissions have a nearly 1:1 relationship (as of 2016). This is strong evidence that there is typically a trade-off between emitting less CO₂ and having a higher GDP. However, Kompas et al. (2018) actually models global warming's long run effect on GDP, and they find that a 3°C rise in global warming causes negative percent change in GDP for almost every country. This challenges the assumption that prioritizing the environment will harm economic output, as these results show how ignoring environmental effects will harm GDP in the long run. There are also additional co-benefits for nations that reduce their CO₂ output, as explored by Liu et al. (2020). For instance, reducing coal and mining plants helps reduce pollutants and fine particulate matter, improving health; reducing automobile use helps reduce traffic congestion, air pollution, accident risk, and damages to roads. Their research found that co-benefits outweigh any associated costs. In regards to the Paris Agreement, they modeled the outcomes for countries that leave the agreement and found the reduction in co-benefits would leave the exiting country worse off, no matter how large it was. In terms of how spread out the effects of climate change are, Newell et al. (2021) state that the negative effect of temperature on GDP levels is especially marked in poorer countries and for agricultural GDP. These results tell us that the effects of climate change are disproportionately felt by poorer countries, which is an important area for UN policy consideration. This conclusion is also supported by Dell et al. (2008), as they find robust

³ *BBC News*. “What Is the Paris Climate Agreement and Why Has Trump Withdrawn?” Science & Environment. December 12, 2015. <https://www.bbc.com/news/science-environment-35073297>.

effects of climate change on poor countries; a 1°C rise in temperature can reduce growth by an average of 1.1 percentage points. This is further exaggerated when each year's effects are compounded, which can ultimately lead to widening the global wealth inequality.

In our research, we intend to analyze the interactions between CO₂ emissions per capita, GDP growth per capita, the UN's Human Development Index (HDI), and the OECD's Environmental Policy Stringency Index (EPS) from 2003-2020. This expands upon further research by exploring the interaction of GDP with different indices, and by connecting these analyses to international trade. We will be testing the following hypotheses: (1) Countries with a higher OECD Environmental Policy Stringency Index can achieve positive GDP per capita growth, controlling for population, CO₂ emissions per capita, HDI, and trade as a percent of GDP. (2) Countries with a higher OECD Environmental Policy Stringency Index can achieve positive HDI levels, controlling for population, CO₂ emissions per capita, GDP growth per capita, and trade as a percent of GDP. (3) Countries whose trade makes up a higher portion of their GDP produce more CO₂ emissions per capita, controlling for population, HDI, and GDP growth per capita.

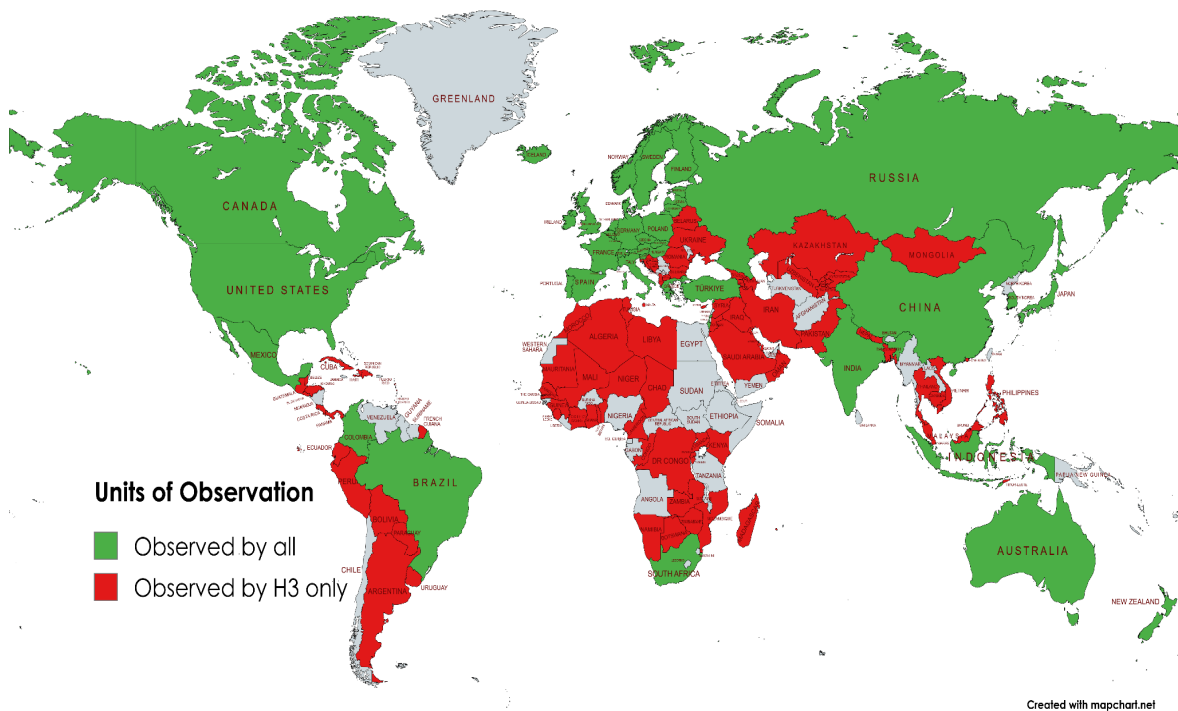
Our topic of inquiry is linked to international trade because it questions the traditional assumption that trade is inherently beneficial to economic growth. While that may be true for a set time, if the long-run externality turns out to be global warming and the exhaustion of natural resources, we must question whether the advantages of increasingly global trade are worth the environmental damage and the financial cost to mitigate climate change. Trade policies like nearshoring have been proposed to alleviate the amount of emissions produced by trade, but implementation has been limited.

Below we outline our data sources, methodology, and econometric modeling. We also analyze our results and interpret them as they relate to our hypotheses. Additionally, we discuss what new developments and policies have been implemented, such as at the COP 30

in Belem, the most recent climate conference. Finally, we discuss the policy implications, shortcomings of our project, and possible future research.

2. Data: Sources, Modeling, & Panel Structure

Panel Structure				
Hypothesis	N (Countries)	T	Format	Model
H1	40	2003-2020	Long & Annual	Fixed effects Panel Regression with distributed lags
H2	40	2003-2020	Long & Annual	Fixed effects Panel Regression with distributed lags
H3	138	2000-2020	Long & Annual	Fixed effects panel regression



Regression Equations:

$$\begin{aligned}
GDP_Growth_{it} &= \beta_0 + \beta_1 EPS_{it} + \beta_2 EPS_{it-1} + \beta_3 EPS_{it-2} + \beta_4 EPS_{it-3} + \\
&\beta_5 \ln(Population)_{it} + \beta_6 CO_2_per_capita_{it} + \beta_7 Trade_pct_GDP_{it} + \beta_8 HDI_{it} + \alpha_i + \varepsilon_{it} \\
HDI_{it} &= \gamma_0 + \gamma_1 EPS_{it} + \gamma_2 EPS_{it-1} + \gamma_3 EPS_{it-2} + \gamma_4 EPS_{it-3} + \gamma_5 \ln(Population)_{it} + \\
&\gamma_6 CO_2_per_capita_{it} + \gamma_7 Trade_pct_GDP_{it} + \gamma_8 GDP_Growth_{it} + \alpha_i + u_{it}^4 \\
CO_2_per_capita_{it} &= \delta_0 + \delta_1 Trade_pct_GDP_{it} + \delta_2 \ln(Population)_{it} + \delta_3 Trade_pct_GDP_{it} + \\
&\alpha_i + v_{it}^5
\end{aligned}$$

We only analyze 40 countries in the first two hypotheses because measures of environmental policy stringency have yet to be measured globally. The time frame is slightly shorter than in the third hypothesis as we use a fixed effects panel regression with distributed lags, common in environmental economics, since the effects of environmental policy differ in the short and medium term. This model combines cross-sectional (40 countries) and time-series (18 years) data. The fixed effect controls for all time-invariant country characteristics (geography, culture, institutions, historical factors) while the within transformation aspect is such that the model estimates effects using within-country variation over time only, removing between-country differences. The distributed lag structure includes current and 3 lagged values of EPS to capture dynamic effects. It is limited in that we lose between-country variation, so we can't estimate effects of time-invariant variables.

We expanded our scope in the third hypothesis, where we observed every country that had data on trade percentages available for the time frame we studied. In this case, we use a fixed effects panel regression.

⁴ Where i = country index (1, 2, ..., 40), t = time period (year: 2003-2020, after losing first 3 years to lags), α_i = country-specific fixed effect (time-invariant country characteristics), u_{it} and ε_{it} = error term.

⁵ Where α_i = country fixed effects and v_{it} = error term

In our analysis, we use country level data from the World Bank on CO₂ emissions per capita,⁶ GDP growth per capita,⁷ trade as % of GDP,⁸ and population.⁹ In addition, we use data from the Organization for Economic Co-operation and Development (OECD)¹⁰, a country-specific and internationally comparable measure of the stringency of environmental policy ranging from zero (least stringent) to six (most stringent). The index covers 40 countries and 13 policy instruments, focusing on climate change and air pollution mitigation policies. The structure of the EPS, its components and the index construction can be accessed in the related 2022 OECD working paper.¹¹ Finally, we use the United Nations' Human Development Index (HDI),¹² a summary measure of achievements in three key dimensions of human development: a long and healthy life, access to knowledge and a decent standard of living. The HDI is on a scale of 0 to 1, it is the geometric mean of normalized indices for each of the three dimensions.

⁶ World Bank. 2024. "Carbon dioxide (CO₂) emissions excluding LULUCF per capita (t CO₂e/capita)." World Bank Open Data. <https://data.worldbank.org>.

⁷ World Bank. 2024. "GDP per capita growth (annual %)." World Bank Open Data. <https://data.worldbank.org>.

⁸ World Bank. 2024. "Trade (% of GDP)." World Bank Open Data. <https://data.worldbank.org>.

⁹ World Bank. 2024. "Population, total." World Bank Open Data. <https://data.worldbank.org>.

¹⁰ Organisation for Economic Co-operation and Development. "OECD Environmental Policy Stringency Index." OECD Publishing, 2024, <https://data.oecd.org>.

¹¹ Kruse, T. et al. (2022). "Measuring environmental policy stringency in OECD countries: An update of the OECD composite EPS indicator", OECD Economics Department Working Papers, No. 1703, OECD Publishing, Paris, <https://doi.org/10.1787/90ab82e8-en>.

¹² United Nations Development Programme 2024. "Human Development Index". <http://hdr.undp.org/en/composite/HDI>

Consolidated Regression Results			
Variable	H1: GDP Growth	H2: HDI	H3: CO ₂ Emissions per Capita
EPS (t)	-1.649*** (0.485)	0.005* (0.002)	—
EPS (t-1)	0.587 (0.597)	0.002 (0.002)	—
EPS (t-2)	0.448 (0.591)	0.001 (0.003)	—
EPS (t-3)	-0.971* (0.450)	0.012*** (0.002)	—
log(Population)	1.453 (3.564)	0.220*** (0.014)	-3.007*** (0.205)
CO ₂ _per_capita	0.516*** (0.117)	0.005*** (0.000)	—
Trade_pct_GDP	0.086*** (0.010)	0.000** (0.000)	-0.013*** (0.001)
HDI	-15.151+ (8.526)	—	10.016*** (0.740)
GDP_Growth	—	-0.000+ (0.000)	0.020*** (0.004)
R ²	0.188	0.692	0.104
Num. Obs.	720	720	2898
+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$			

3. Model Analysis & Figures

Interpreting the coefficients of hypothesis 1's regression, we find that a one-unit increase in a country's Environmental Policy Stringency Index is associated with a 1.649% decrease in GDP per capita growth that same year. This reflects immediate economic adjustment costs, where firms might have to invest in cleaner technology, comply with regulations, reduce output, and face higher operational costs. The effect after 1 or 2 years is positive (0.587% & 0.448%) and suggests a recovery, but it is not statistically distinguishable

from 0 given their respective standard errors (0.597% & 0.448%). After 3 years, gdp growth decreases by 0.971%, so the negative persists over the medium-term. The cumulative effect of a 1 point increase in EPS yields a cumulative decrease in GDP growth of 2.585% over 4 years. Hypothesis 1 fails, but with low explanatory power ($R^2 = 0.188$).

Figure 1.1

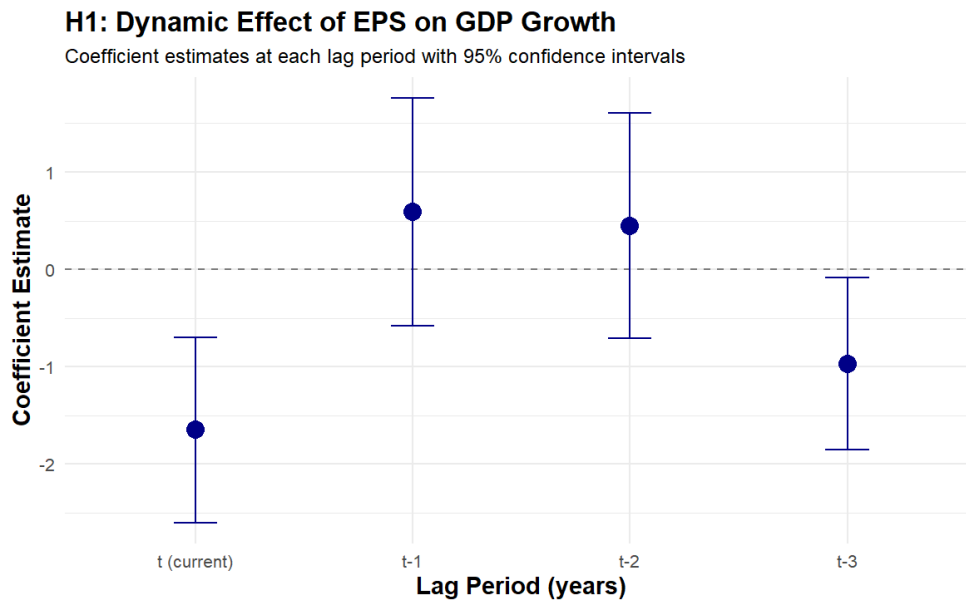
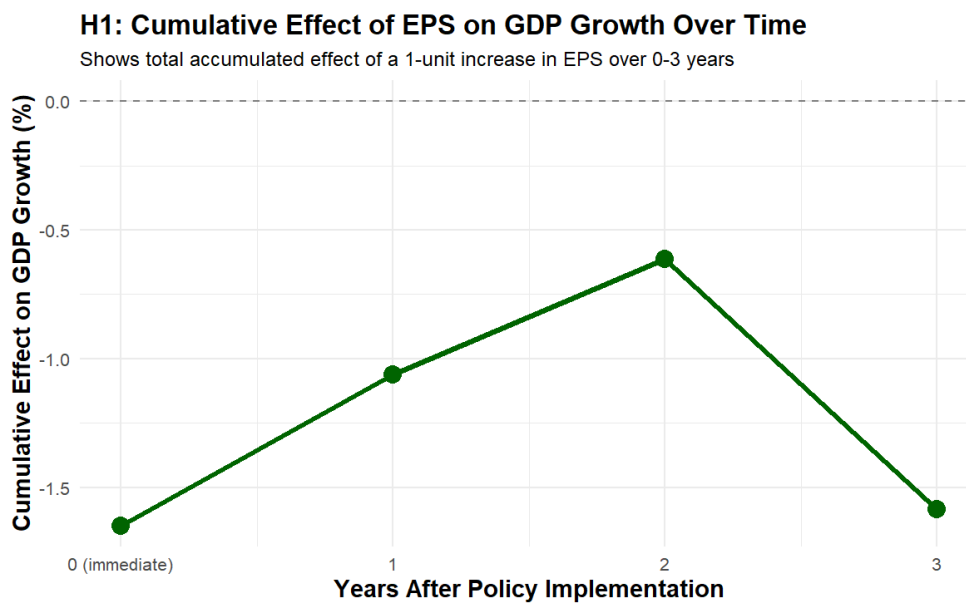


Figure 1.2



Interpreting the coefficients of hypothesis 2's regression, we find that a one-unit increase in a country's Environmental Policy Stringency Index is associated with a 0.005 increase in Human Development Index (index with range 0-1) that same year. It could be that environmental policies immediately improve quality of life through cleaner air, better health outcomes, and reduced pollution-related mortality. The effect after 1 or 2 years is also positive (0.002 & 0.001) but not statistically distinguishable from 0 given their standard errors (0.002 & 0.003). After 3 years, HDI increases by 0.012 which is the strongest effect and more than double the immediate one. The cumulative effect of a 1 point increase in EPS yields a cumulative increase in HDI of 0.02 over 4 years (on a scale of 1), suggesting that the benefits to human development compound and strengthen over time. These benefits might arise from job creation in green industries, accumulated health benefits, and improved quality of life metrics (reduced air pollution, life expectancy, green spaces, etc). Hypothesis 2 succeeds with high explanatory power ($R^2 = 0.692$).

Figure 2.1

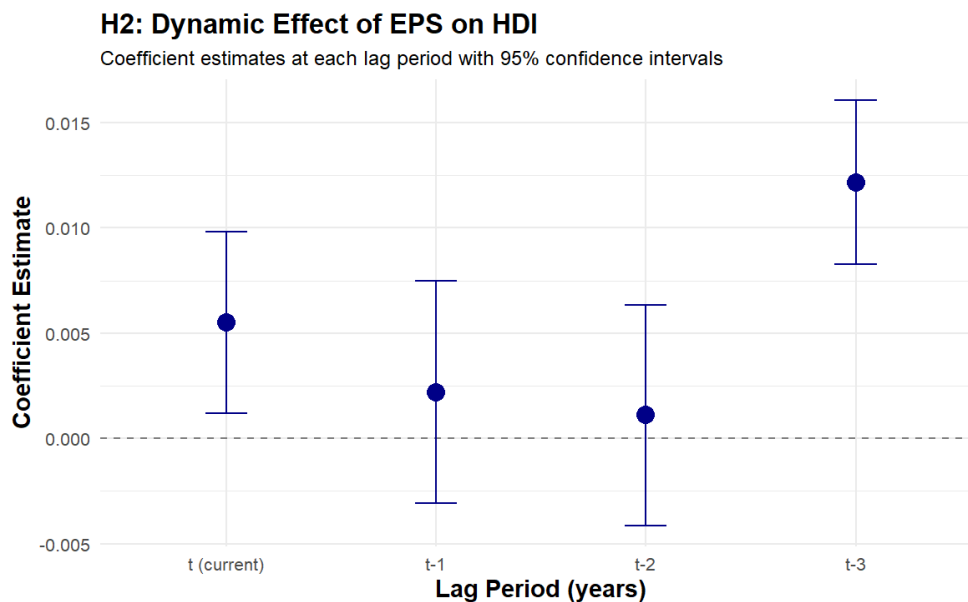
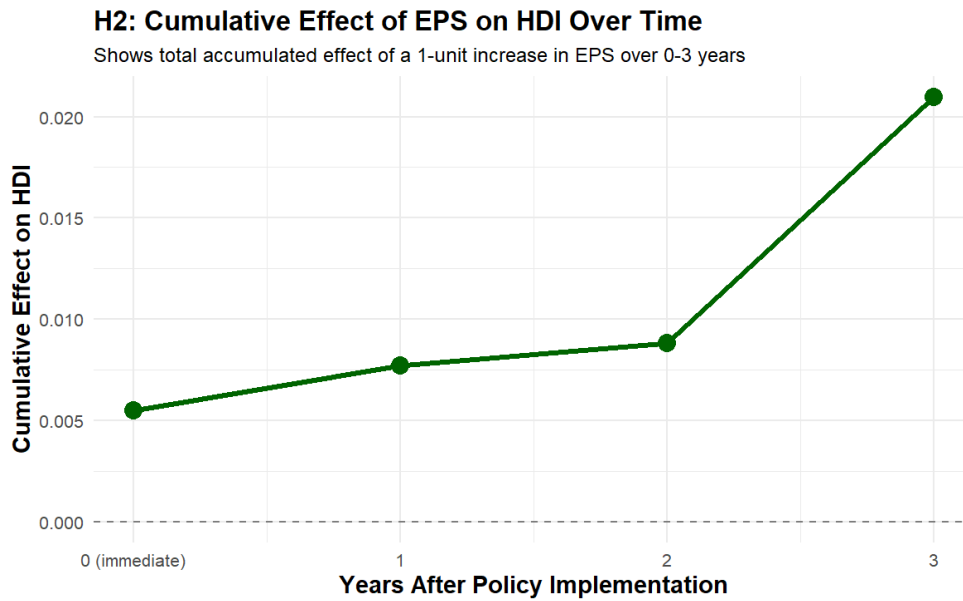
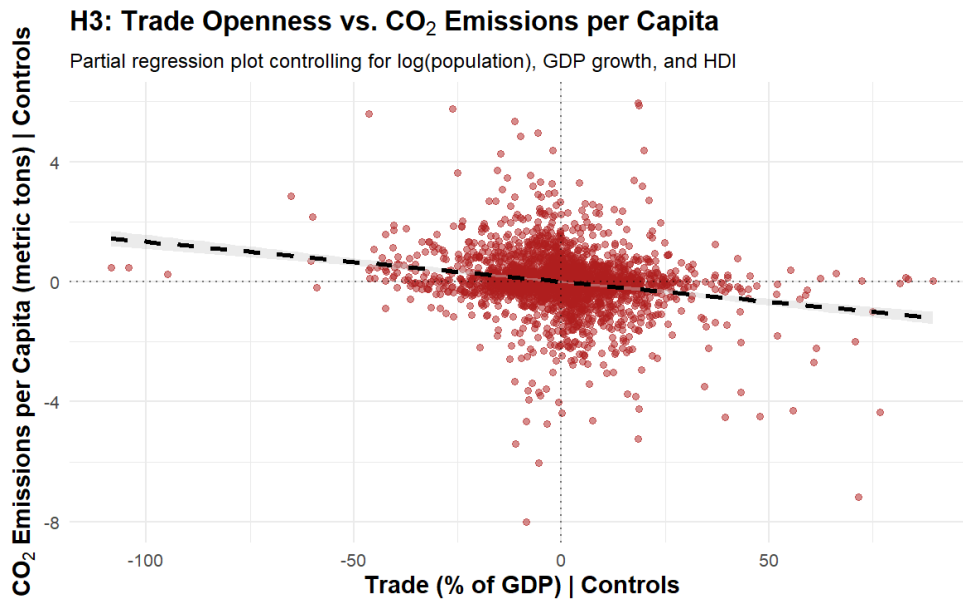


Figure 2.2

Interpreting the coefficients of hypothesis 3's regression, we find that a one-percentage point increase in a country's trade as % of GDP is associated with a 0.013 metric tons decrease in CO₂ emissions per capita. One possible explanation for why trade might reduce emissions is that countries specialize in what they do effectively, so labor-abundant countries might specialize in less emission-intensive sectors (services, light manufacturing) while capital-rich countries handle heavy industry more efficiently. Trade could also facilitate the transfer of clean technologies. Hypothesis 3 fails, but with low explanatory power ($R^2 = 0.104$). This is likely because CO₂ emissions are influenced by many factors not captured here like energy mix, industrial structure, climate, technology, natural resource endowments, and offshoring.

Figure 3

4. Policy Implications & Discussion

The Cop 30 recently held in Belém, Brazil¹³ hosted 190 countries with the notable exception of the US to advance the aforementioned Paris agreement. Major polluters and petrostates (mainly Russia, Saudi Arabia, & India) had significant pull in the negotiations, resulting in the absence of a fossil fuels mention in the conference’s primary agreement. According to the Global Witness organization, 1/25 attendees were fossil fuel lobbyists, the highest proportion of any COP. Climate adaptation finance promises nonetheless tripled, and signatories agreed to raise \$1.3 trillion annually for climate action by 2035. While outcomes of climate agreements continue to fall short of the mitigation policies required to stabilize global warming, our research points to a nuanced but optimistic picture of sustainable development. When looking at the short to medium term, environmental policy enforcement

¹³ Global Witness. “COP30 Verdict: People in, Polluters Still In?”
https://globalwitness.org/en/campaigns/fossil-fuels/cop30-verdict-people-in-polluters-still-in/?gad_source=1&gad_campaignid=12467226173&gbraid=0AAAAADm6LODH6eBgdxEQtfH6sXv18ISfc.

negatively affected GDP growth per capita. However, this effect was positive on Human Development Index levels, which encompasses a larger breadth of economic outcomes than GDP growth. The explanatory power (R^2) of our GDP growth findings was also more limited than that of our HDI findings, where more of the variation was explained by Environmental policy stringency. Despite the alarm bells rung by critics of sustainable policy enforcement, the balance between economic development and protecting the environment does not have to be a zero sum game.

5. Conclusion

Our paper was motivated by worsening climate change and the escalating urgency to address it. Multilateral policy that enforces environmental sustainability is a prerequisite to widespread change, shifting the responsibility away from the individual consumer.¹⁴ In our research, we sought to explore the validity of the relationship between environmental sustainability and GDP. Further, we compared this relationship to a different measure, the Human Development Index. Finally, we analyzed CO₂ emissions and their link to trade. Our main results confirm that increased environmental policy stringency leads to a decrease in a country's GDP per capita growth in the short to medium term. Conversely, we found that increased environmental policy stringency leads to an increase in a country's human development, an alternate measure of a country's well-being. Finally, our regression shows that an increase in a country's trade (as a percentage of their GDP) leads to a decrease in CO₂ emissions.

The first limitation we ran into during our research was data availability for our specific variables. Only 40 countries had data on Environmental Policy Stringency in our

¹⁴ Dana R Fisher, "Taking Power as Individuals (and Why Individual Climate Action Can't Save Us)," Brookings, February 13, 2024, <https://www.brookings.edu/articles/taking-power-as-individuals-and-why-individual-climate-action-cant-save-us/>.

timeframe. Because we had to exclude many nations (notably most South American and African countries), it is difficult to draw universal conclusions, especially given the underrepresented regions. It is important to note this when interpreting our analyses, because our results may be more heavily representative of developed countries. An area worthy of future research could be to investigate the environmental policy stringency of developing countries once more data becomes available (or using a different metric than EPS), and explore how the effects on economic development differ in those regions. Additionally, adding different dependent variables such as income inequality, poverty, and health outcomes could provide valuable insight.

Another limitation of our research stems from our analysis of GDP and HDI. We analyzed GDP in terms of percent growth, while HDI was measured as levels. In future research, choosing to analyze these effects uniformly in percentage terms or levels would help us draw more concrete conclusions, instead of mixing the two. Our final recommendation for future research would be to create a predictive model for long-term effects. Since we are only able to analyze past trends, our data only allows for short-medium term effects as the magnitude of climate change has yet to be seen. Modeling the future climate and growth effects would help us visualize outcomes from meeting climate deliverables and implementing sustainable policy.

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